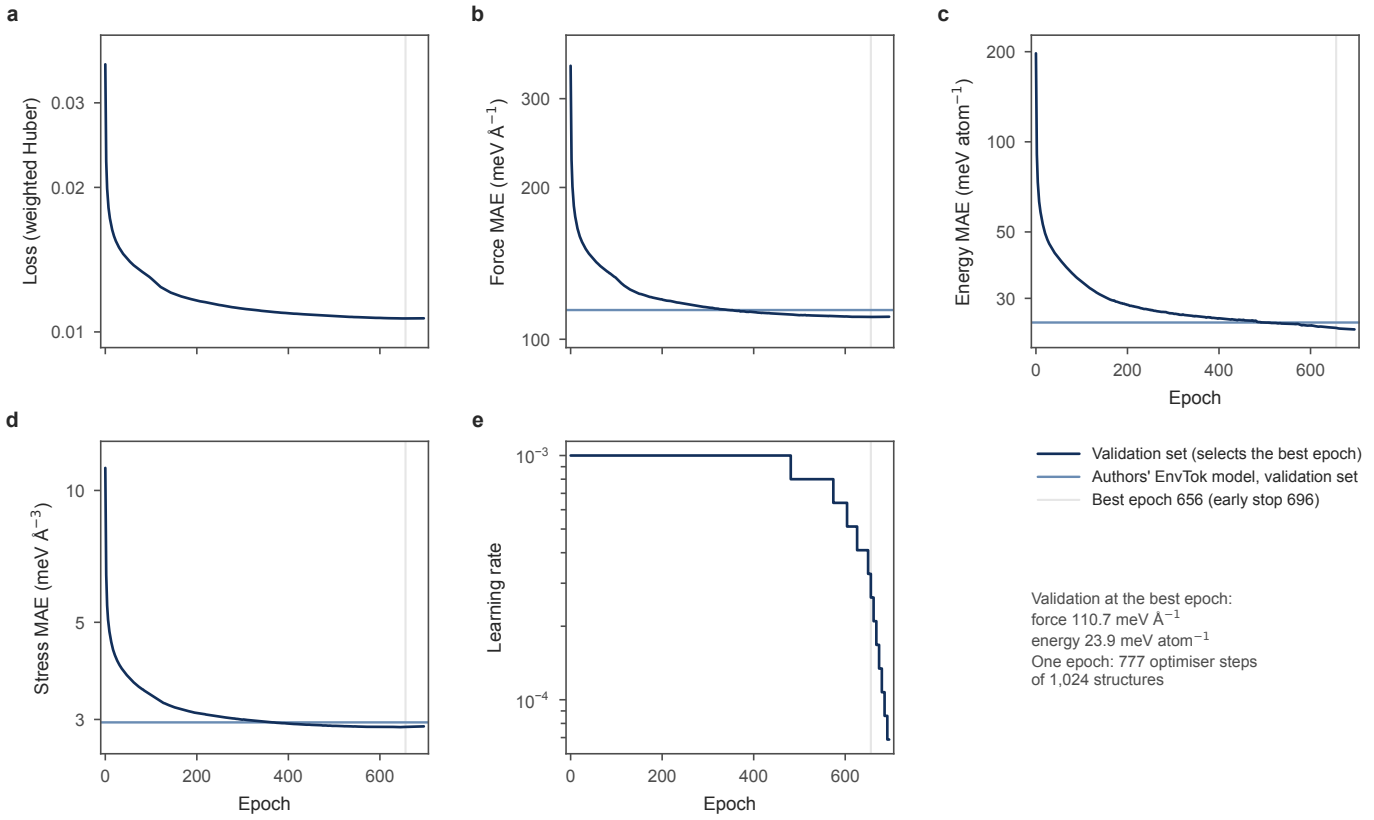
a
Loss (weighted Huber)
0.03
0.02
0.01
b
Force MAE (meV Å^{-1})
300
200
100
c
Energy MAE (meV atom^{-1})
200
100
50
30
0
200
400
600
Epoch
d
Stress MAE (meV Å^{-3})
10
5
3
0
200
400
600
Epoch
e
Learning rate
10^{-3}
10^{-4}
0
200
400
600
Epoch
Validation set (selects the best epoch)
Authors' EnvTok model, validation set
Best epoch 656 (early stop 696)
Validation at the best epoch:
force 110.7 meV Å^{-1}
energy 23.9 meV atom^{-1}
One epoch: 777 optimiser steps
of 1,024 structures